Analyze E-Commerce Sales to Improve Customer Conversion and Retention

1. Introduction:

The e-commerce industry generates large volumes of customer and transaction data that can be leveraged to improve business performance. This project, **"Analyze E-Commerce Sales to Improve Customer Conversion and Retention,"** focuses on examining customer purchasing behaviour, sales trends, product performance, and retention patterns using data analytics techniques.

Using **Python, SQL, Excel, and Power BI**, the project performs data cleaning, exploratory data analysis (EDA), dashboard development, statistical validation, and regression analysis on e-commerce datasets. The analysis aims to identify factors influencing sales growth, customer conversion, repeat purchases, and customer satisfaction.

Key performance indicators such as **Total Revenue, Average Order Value (AOV), Repeat Purchase Rate, Review Score, Cancellation Rate, and Delivery Performance** are analyzed to evaluate business performance. The insights generated help identify profitable customer segments, high-performing product categories, and opportunities to improve customer retention and operational efficiency.

The findings provide data-driven recommendations that support better decision-making and contribute to sustainable business growth in the e-commerce sector.

2. Problem Statement:

E-commerce businesses generate vast amounts of customer and transaction data but often lack clear insights into the factors affecting customer conversion, retention, and revenue growth. This project analyzes sales, customer, product, and delivery data to identify key business drivers and provide actionable recommendations for improving customer loyalty, sales performance, and profitability.

3. Business Objective:

Objective 1: To identify category- and segment-level drivers of first purchase conversion and overall sales growth

Objective 2: To understand repeat purchase, volume, and loyalty patterns across demographics, categories, and delivery performance

Objective 3: To optimize product portfolio, pricing, discounting, and shipping for profitable growth

Objective 4: To leverage reviews, ratings, and customer sentiment to enhance conversion and retention

Objective 5: To evaluate logistics and operational efficiency to improve customer experience

Objective 6: To optimize customer retention and lifetime value across key product categories

4. Business Objective wise Questions:

Objective 1: To identify category- and segment-level drivers of first purchase conversion and overall sales growth

- **Question 1:** Which product categories and subcategories have the highest add-to-cart to purchase conversion and lowest cancellation rates?
- **Question 2:** How do conversion rates vary by customer segment (New vs. Loyal), gender, and age group?
- **Question 3:** How do sales revenues vary across different months?
- **Question 4:** How has overall sales performance changed over the years?

Objective 2: To understand repeat purchase, volume, and loyalty patterns across demographics, categories, and delivery performance

- **Question 1:** Which categories and brands most strongly predict repeat purchases among New customers transitioning to Loyal?
- **Question 2:** How do discount rates, shipping costs, and unit prices affect reorder probability within 30/60/90 days?
- **Question 3:** Which Categories Have the Top 10 Highest Order Volume?
- **Question 4:** What is the Repeat Purchase Rate?

Objective 3: To optimize product portfolio, pricing, discounting, and shipping for profitable growth

- **Question 1:** What is item-level margin after discount and shipping, by category, brand, and stock availability?
- **Question 2:** How do changes in discount bands influence conversion and average order value across segments?
- **Question 3:** Which Product Categories Generate the Top 10 Highest Revenue?
- **Question 4:** How does product pricing influence customer satisfaction and review ratings?

Objective 4: To leverage reviews, ratings, and customer sentiment to enhance conversion and retention

- **Question 1:** Which categories and brands consistently earn high ratings and low returns/cancellations?
- **Question 2:** How do review scores correlate with reorder rates and time-to-next-purchase?
- **Question 3:** Which Product Categories Receive the Best Reviews?
- **Question 4:** How do positive vs. negative review trends impact overall category conversion rates?

Objective 5: To evaluate logistics and operational efficiency to improve customer experience

- **Question 1:** How Does Delivery Performance Affect Customer Satisfaction?
- **Question 2:** How do shipping costs and delivery times vary across product categories and customer regions?
- **Question 3:** What impact do delivery delays have on order cancellation and product return rates?
- **Question 4:** How does fulfillment speed affect customer review ratings and overall satisfaction scores?

Objective 6: To optimize customer retention and lifetime value across key product categories

- **Question 1:** What are the primary drivers of customer churn across high-volume product categories?

- **Question 2:** How does the time elapsed between initial and second purchases differ across product categories?
- **Question 3:** Which customer segments generate the highest total revenue over their purchasing lifetime?
- **Question 4:** How do targeted discounts and loyalty incentives impact re-engagement rates for inactive buyers?

5. Data Description:

This project utilizes cleaned e-commerce transaction data collected from an online retail platform to analyze customer conversion, retention, sales performance, and operational efficiency. The data belongs to the e-commerce domain and captures the complete customer purchase lifecycle, including customer information, order transactions, product details, reviews, and geographical attributes.

The analysis was performed using six integrated datasets: Customers, Orders, Order Items, Products, Reviews, and Geolocation. These datasets were merged using common identifiers such as Customer ID, Order ID, and Product ID to create a unified analytical dataset. The integrated dataset provides a comprehensive view of customer behaviour, purchasing patterns, product performance, delivery outcomes, and customer satisfaction.

The **Customers dataset** contains customer demographic and segmentation information, including customer ID, gender, age group, state, and customer segment (New or Loyal). The **Orders dataset** stores order-level details such as order ID, customer ID, order status, purchase timestamp, estimated delivery date, and actual delivery date. The **Order Items dataset** contains transaction-level information including product ID, quantity, unit price, freight charges, discounts, and revenue-related attributes. The **Products dataset** provides product category and subcategory information used for category-wise sales and profitability analysis. The **Reviews dataset** contains customer ratings and feedback used to measure customer satisfaction and service quality. The **Geolocation dataset** includes location-based attributes such as city, state, and zip code that support geographical analysis.

Several derived variables were created during data preparation, including Revenue, Average Order Value (AOV), Delivery Delay Days, Repeat Purchase Indicator, Profit, Profit Margin, Purchase Month, Purchase Year, Customer Lifetime Value indicators, and Delivery Status. These variables were used throughout the project for exploratory data analysis, SQL-based business analysis, dashboard development, statistical hypothesis testing, and regression modelling.

The final cleaned dataset served as the foundation for evaluating key business metrics such as Revenue, Order Volume, Repeat Purchase Rate, Customer Satisfaction Score, Cancellation Rate, Delivery Performance, and Customer Retention, enabling data-driven insights and recommendations for improving customer conversion and retention.

6. Technology Stack:

The project was executed using a combination of data analytics, visualization, statistical analysis, and machine learning tools to transform raw e-commerce data into actionable business insights.

Microsoft Excel:

Microsoft Excel was used for preliminary data validation, KPI calculations, pivot table analysis, and dashboard development. It enabled quick exploration of business metrics such as Revenue, Average Order Value (AOV), Repeat Purchase Rate, Review Score, and Delivery Performance through interactive visualizations and slicers.

SQL:

SQL was used for data extraction, transformation, validation, and business analysis. Complex queries were written to identify sales trends, customer behavior patterns, category performance, order status distribution, and profitability metrics. A consolidated master view was created to integrate multiple datasets for analysis.

Python:

Python was used for data cleaning, preprocessing, exploratory data analysis (EDA), statistical testing, and regression modeling. Libraries such as Pandas, NumPy, Matplotlib, and Seaborn were utilized to analyze customer behavior, sales performance, delivery efficiency, and retention patterns.

Power BI:

Power BI was used to develop interactive dashboards and business intelligence reports. It enabled dynamic visualization of key performance indicators (KPIs), sales trends, customer segmentation, category performance, and operational metrics, supporting data-driven decision-making.

Statistical Analysis:

Statistical hypothesis testing techniques such as Z-tests and T-tests were applied to validate business assumptions related to customer retention, delivery performance, profitability, and customer satisfaction. These tests helped determine whether observed differences were statistically significant.

Machine Learning (Linear Regression):

Linear Regression models were developed to examine relationships between business variables such as delivery delay, fulfillment time, unit price, customer reviews, and churn behavior. The models provided predictive insights and helped identify factors influencing customer satisfaction and retention.

7. Data Cleaning & Fixes Applied:

Data quality assessment was performed before analysis to ensure accuracy, consistency, and reliability of the results. Multiple inconsistencies and discrepancies were identified across the Customers, Orders, Order Items, Reviews, Products, and Geolocation datasets. Appropriate data cleaning techniques were applied to address these issues.

Missing Values Handling

Several datasets contained missing or incomplete records. Missing values were identified using null-value analysis and handled appropriately based on business relevance. Critical fields required for analysis were retained, while non-essential missing values were either imputed or excluded from specific analyses to prevent biased results.

Duplicate Record Removal

Duplicate records were checked across all datasets using unique identifiers such as Customer ID, Order ID, Product ID, and Review ID. Duplicate entries were removed to avoid double-counting of transactions, customers, or revenue figures.

Data Type Corrections

Certain columns contained incorrect data formats. Date fields were converted to proper datetime formats to enable time-series analysis, while numerical columns such as prices, freight charges, discounts, and review scores were converted to appropriate numeric data types for accurate calculations.

Standardization of Categorical Variables

Categorical fields such as Order Status, Customer Segment, Product Category, Gender, and Delivery Status were standardized by correcting inconsistent spellings, formatting differences, and case sensitivity issues. This ensured uniform categorization across analyses and dashboards.

Validation of Business Rules

Logical validation checks were performed to identify invalid or inconsistent records. Examples included:

- Orders with missing customer identifiers.
- Negative or invalid price values.
- Inconsistent delivery dates.
- Incorrect order status classifications.
- Review scores outside the expected range.

Records violating business rules were corrected where possible or excluded from analytical calculations.

Date and Delivery Calculations

Date-related columns were cleaned and transformed to derive business metrics such as:

- Delivery Delay Days
- Fulfillment Time

- Purchase Month
- Purchase Year
- Delivery Status (On-Time / Delayed)

These derived variables were used extensively in dashboarding, statistical analysis, and regression modeling.

Creation of Derived Business Metrics

Additional analytical variables were generated to support business objectives, including:

- Revenue
- Profit
- Profit Margin
- Average Order Value (AOV)
- Repeat Purchase Indicator
- Customer Retention Metrics
- Cancellation Metrics

Data Integration

The Customers, Orders, Order Items, Products, Reviews, and Geolocation datasets were merged using common keys such as Customer ID, Order ID, and Product ID. The integrated dataset was validated to ensure referential integrity and consistency across all records.

Outcome of Data Cleaning

After applying these cleaning and transformation steps, a consolidated and analysis-ready dataset was created. This dataset served as the foundation for SQL analysis, exploratory data analysis (EDA), dashboard development, hypothesis testing, and machine learning models used throughout the project.

8. Tools and Techniques Used:

The successful completion of this capstone project required the application of various analytical tools and techniques to transform raw e-commerce data into actionable business insights.

Tools Used

Microsoft Excel

- Data validation and preliminary analysis
- KPI calculations
- Pivot Tables and Pivot Charts
- Interactive Dashboard Development
- Slicers and Filters for dynamic reporting

SQL

- Data extraction and transformation
- Data quality validation

- Dataset integration through joins
- Creation of analytical views
- Business performance analysis using aggregation and grouping techniques

Python

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Statistical hypothesis testing
- Regression modeling
- Data visualization

Power BI

- Interactive dashboard development
- KPI monitoring
- Trend analysis and business reporting
- Data visualization and storytelling

Techniques Used

Data Cleaning and Preprocessing

- Missing value identification and treatment
- Duplicate record validation
- Data type conversion
- Standardization of categorical variables
- Feature creation and transformation

Data Integration

- Dataset merging using common keys
- Creation of a consolidated analytical dataset
- Validation of referential integrity

Exploratory Data Analysis (EDA)

- Univariate analysis
- Bivariate analysis
- Trend analysis
- Category-wise analysis
- Customer segmentation analysis

Descriptive Analytics

- Revenue analysis
- Order volume analysis
- Customer behavior analysis
- Product performance analysis
- Delivery performance analysis

KPI Analysis

- Revenue
- Average Order Value (AOV)
- Repeat Purchase Rate
- Cancellation Rate
- Review Score Analysis
- Delivery Performance Metrics

Statistical Analysis

- Descriptive Statistics
- Hypothesis Testing
- Two-Proportion Z-Test
- Independent Sample T-Test
- Statistical Significance Testing using p-values

Data Visualization

- Bar Charts
- Line Charts
- Pie Charts
- Column Charts
- Scatter Plot
- Dashboard Visualizations

Machine Learning

- Feature Engineering
- Train-Test Split
- Simple Linear Regression
- Model Evaluation

Model Evaluation Techniques

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Regression Coefficient Interpretation

The combination of these tools and techniques enabled comprehensive analysis of customer conversion, retention behavior, sales performance, operational efficiency, and customer satisfaction, ultimately supporting data-driven business decision-making.

9. Possible Analysis:

To achieve the project objectives of improving customer conversion, retention, and sales performance, multiple types of analyses were performed on the e-commerce dataset. Each analysis provides unique business insights and supports data-driven decision-making.

Descriptive Analysis

Descriptive analysis was conducted to summarize historical business performance and understand the current state of the business.

Analysis Performed:

- Total Revenue Analysis
- Total Orders Analysis
- Average Order Value (AOV)
- Repeat Purchase Rate
- Cancellation Rate
- Review Score Analysis
- Delivery Performance Analysis

Business Value:
Provides a high-level understanding of business performance and identifies key trends and patterns.

Customer Analysis

Customer analysis was performed to understand purchasing behaviour, customer segments, and retention patterns.

Analysis Performed:

- New vs Loyal Customer Analysis
- Repeat Purchase Analysis
- Customer Revenue Contribution
- Customer Segment Performance
- Customer Satisfaction Analysis

Business Value:
Helps identify valuable customer segments and develop effective retention strategies.

Product and Category Analysis

Product-level analysis was conducted to evaluate category performance and identify high-performing products.

Analysis Performed:

- Revenue by Product Category
- Order Volume by Category
- Top Revenue-Generating Products
- Category Profitability Analysis
- Product Performance Comparison

Business Value:
Supports product portfolio optimization and inventory planning.

Sales Trend Analysis

Sales trend analysis was used to identify temporal patterns in revenue and order volume.

Analysis Performed:

- Monthly Sales Trends
- Yearly Sales Trends
- Seasonal Revenue Analysis
- Order Volume Trends

Business Value:
Helps forecast demand and identify peak sales periods.

Operational Performance Analysis

Operational analysis was conducted to assess delivery efficiency and order fulfillment performance.

Analysis Performed:

- Delivery Delay Analysis
- On-Time Delivery Performance
- Order Status Analysis
- Cancellation Analysis
- Fulfillment Time Analysis

Business Value:
Identifies operational bottlenecks that affect customer experience and retention.

Customer Satisfaction Analysis

Customer review and rating data were analyzed to understand service quality and customer sentiment.

Analysis Performed:

- Review Score Distribution
- Review Score by Delivery Status
- Review Score by Product Category
- Relationship Between Delivery Performance and Reviews

Business Value:
Helps improve customer experience and service quality.

Statistical Analysis

Statistical techniques were applied to validate business assumptions and determine whether observed differences were statistically significant.

Analysis Performed:

- Descriptive Statistics
- Two-Proportion Z-Test
- Independent Sample T-Test
- Hypothesis Validation

Business Value:
Provides evidence-based decision-making rather than relying solely on observations.

Correlation and Relationship Analysis

Relationship analysis was performed to understand the impact of one business variable on another.

Analysis Performed:

- Price vs Review Score
- Delivery Delay vs Review Score
- Fulfillment Time vs Customer Satisfaction
- Repeat Purchase vs Revenue

Business Value:
Identifies the factors that influence customer satisfaction and retention.

Predictive Analysis (Machine Learning)

Regression modeling was performed to predict business outcomes and understand key influencing factors.

Analysis Performed:

- Linear Regression Modeling
- Delivery Delay Impact Prediction
- Fulfillment Time Impact Analysis
- Customer Satisfaction Prediction

Business Value:
Supports proactive decision-making and future business planning.

Dashboard and Business Intelligence Analysis

Interactive dashboards were developed using Excel and Power BI to monitor business performance through key performance indicators.

Analysis Performed:

- KPI Monitoring

- Revenue Dashboard
- Customer Retention Dashboard
- Operational Performance Dashboard

Business Value:
Enables stakeholders to track performance, identify trends, and make timely business decisions through visual analytics.

10.Hypotheses:

Objective 1:

- **Question 1:** Certain product categories and subcategories achieve significantly higher conversion rates and lower cancellation rates than others.
- **Question 2:** Loyal customers and middle-aged customer groups exhibit higher purchase conversion rates compared to new and younger customers.
- **Question 3:** Sales revenue demonstrates seasonal fluctuations, with specific months generating significantly higher revenue than others.
- **Question 4:** Overall sales revenue and order volume have shown a positive growth trend over time.

Objective 2:

- **Question 1:** Customers purchasing from specific categories and brands are more likely to make repeat purchases and become loyal customers.
- **Question 2:** Higher discounts and lower shipping costs increase the likelihood of repeat purchases within 90 days.
- **Question 3:** A small number of product categories account for a disproportionately large share of total order volume.
- **Question 4:** Repeat purchases contribute a substantial portion of total orders, indicating the presence of a loyal customer base.

Objective 3:

- **Question 1:** Profit margins vary significantly across product categories and brands after accounting for discounts and shipping costs.
- **Question 2:** Moderate discount levels improve conversion rates and average order value more effectively than very low or very high discount levels.
- **Question 3:** Revenue generation is concentrated within a limited number of high-performing product categories.
- **Question 4:** Higher-priced products tend to receive better customer ratings due to perceived quality and value.

Objective 4:

- **Question 1:** Products with consistently high ratings experience lower cancellation rates and stronger customer retention.
- **Question 2:** Customers who provide higher review scores are more likely to repurchase and do so within shorter time intervals.
- **Question 3:** Certain product categories consistently achieve higher average review scores than others.

- **Question 4:** Categories with predominantly positive reviews exhibit higher conversion rates than categories with negative review trends.

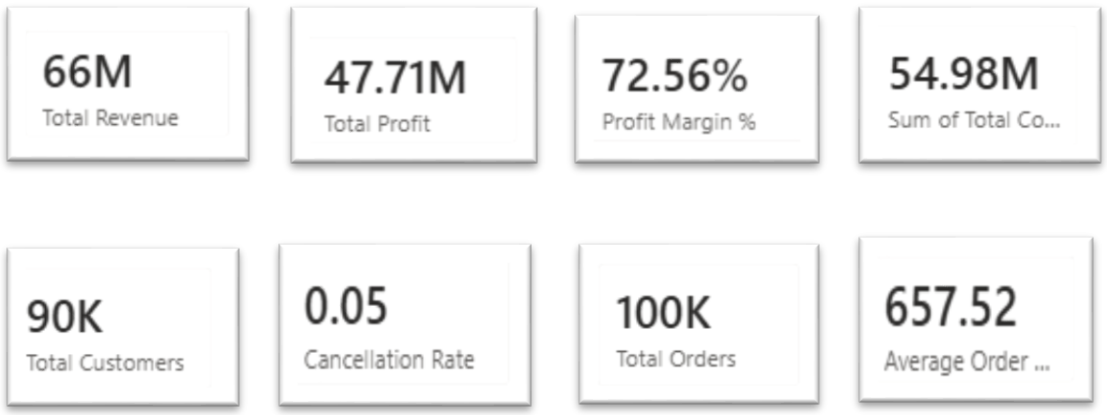
Objective 5:

- **Question 1:** Faster and on-time deliveries are associated with higher customer satisfaction ratings.
- **Question 2:** Shipping costs and delivery times differ significantly across product categories and geographic regions.
- **Question 3:** Orders experiencing delivery delays have higher cancellation and return rates than on-time deliveries.
- **Question 4:** Shorter fulfillment times lead to higher review ratings and overall customer satisfaction.

Objective 6:

- **Question 1:** Low customer ratings, higher prices, and delivery delays are associated with increased customer churn.
- **Question 2:** Customers purchasing from frequently consumed product categories make their second purchase sooner than customers in other categories.
- **Question 3:** Loyal customers contribute significantly higher lifetime revenue than new or occasional customers.
- **Question 4:** Customers receiving discounts and promotional incentives are more likely to return and make subsequent purchases.

11.KPIs:



12.Exploratory Insights:

Sales Insights:

- Monthly sales show seasonal fluctuations with certain months contributing significantly higher revenue.
- Revenue generation is concentrated in a limited number of product categories.
- Sales performance demonstrates variation across different periods, indicating changing customer demand patterns.

Customer Insights:

- Loyal customers contribute substantially more revenue than one-time customers.
- Repeat purchasers tend to have higher average spending compared to new customers.

- Customer retention plays a major role in overall business revenue.

Product Insights:

- A few product categories account for the majority of sales and order volume.
- Product performance varies considerably across categories and subcategories.
- High-performing categories provide opportunities for targeted marketing and inventory optimization.

Delivery Insights:

- Delivery performance influences customer experience and satisfaction.
- Orders delivered on time generally receive better customer ratings.
- Delivery delays can negatively affect customer perception of service quality.

Customer Satisfaction Insights:

- Most customers provide positive review scores, indicating overall satisfaction.
- Review ratings vary across delivery performance levels and product categories.
- Customer feedback provides useful indicators of operational effectiveness.

Business Insights:

- Customer retention contributes more to sustainable growth than relying solely on new customer acquisition.
- Category-level performance differences highlight opportunities for strategic product planning.
- Improving delivery efficiency and customer experience can positively impact repeat purchase behavior.

13. Insights From SQL Scripts:

The SQL analysis was performed to address the project's business objectives related to customer conversion, retention, sales growth, and profitability. By integrating customer, order, product, review, and delivery data into a consolidated analytical view, meaningful business insights were generated.

Business Objective 1:

Identify Category- and Segment-Level Drivers of Customer Conversion and Sales Growth

Key SQL Insights:

- Category-wise sales analysis revealed that revenue generation is concentrated within a few high-performing product categories, indicating their significant contribution to overall business growth.
- Order volume analysis showed that customer demand varies across categories, helping identify products with strong market acceptance.
- Customer segment analysis highlighted differences in purchasing behavior between New and Loyal customers.
- Revenue trend analysis identified monthly and yearly fluctuations in sales performance, indicating seasonal demand patterns and growth opportunities.
- Order status analysis revealed the proportion of completed, cancelled, and unavailable orders, helping assess factors affecting successful conversion.

Business Impact:

These insights help management focus marketing efforts on high-performing categories, improve conversion rates, and allocate resources more effectively.

Business Objective 2:

Understand Customer Retention and Repeat Purchase Behaviour

Key SQL Insights:

- Repeat purchase analysis identified a segment of customers responsible for a significant share of total revenue.
- Loyal customers demonstrated higher purchase frequency and greater revenue contribution than one-time customers.
- Customer-level transaction analysis revealed that repeat customers generate substantially higher business value than newly acquired customers.
- Review and purchase history integration enabled identification of factors associated with customer satisfaction and retention.
- Customer segmentation analysis helped distinguish retention patterns across different customer groups.

Business Impact:

These findings support the development of customer loyalty programs, retention campaigns, and personalized marketing strategies to increase repeat purchases.

Business Objective 3:

Optimize Product Portfolio, Pricing, Delivery, and Profitability

Key SQL Insights:

- Profitability calculations revealed variations in profit margins across product categories and products.
- Freight cost and shipping analysis provided visibility into operational costs affecting overall profitability.
- Delivery performance analysis identified delayed deliveries and their potential impact on customer experience.
- Product-level revenue and profit analysis helped identify products contributing most to business growth.
- The integrated master view enabled comparison of revenue, cost, profit, and delivery metrics at both product and category levels.

Business Impact:

These insights assist management in optimizing pricing strategies, improving delivery performance, reducing operational costs, and maximizing profitability.

Overall SQL Contribution to the Project:

The SQL analysis provided a structured foundation for business decision-making by:

- Identifying high-performing categories and customer segments.
- Measuring customer retention and repeat purchase behaviour.
- Evaluating operational efficiency through delivery and order status analysis.

- Assessing profitability across products and categories.
- Supporting data-driven strategies to improve customer conversion, retention, and long-term business growth.

14.EDA Insights:

Objective 1: Identify Category- and Segment-Level Drivers of First Purchase Conversion and Overall Sales Growth

Key Insights:

- Revenue generation is concentrated within a limited number of product categories, indicating that certain categories play a dominant role in overall business growth.
- Monthly sales analysis revealed noticeable fluctuations in revenue, suggesting the presence of seasonal purchasing patterns.
- Year-wise sales trends indicated positive business growth over time, with increasing order volume and revenue.
- Customer segment analysis showed differences in purchasing behavior between New and Loyal customers.
- Age-group and demographic analysis highlighted variations in purchasing activity across customer groups.

Business Implication:

These insights help identify high-performing categories and customer segments that can be targeted to improve conversion and revenue growth.

Objective 2: Understand Repeat Purchase, Volume, and Loyalty Patterns Across Demographics, Categories, and Delivery Performance

Key Insights:

- Repeat customers contribute significantly more revenue than one-time buyers.
- A small group of customers accounts for a substantial share of repeat purchases and overall revenue.
- Certain product categories demonstrate stronger repeat purchase behavior than others.
- Order volume analysis showed that a limited number of categories account for the majority of customer demand.
- Loyal customers exhibit higher purchasing frequency and greater spending levels compared to New customers.

Business Implication:

The findings highlight the importance of customer retention strategies and category-specific loyalty initiatives.

Objective 3: Optimize Product Portfolio, Pricing, Discounting, and Shipping for Profitable Growth

Key Insights:

- Revenue contribution varies significantly across product categories, indicating differences in category performance.
- Product categories generating the highest revenue are not always the most profitable categories.
- Pricing patterns suggest variation in customer purchasing behavior across different price ranges.
- Shipping costs contribute substantially to overall transaction costs and influence profitability.
- Product-level analysis identified categories with strong revenue potential and categories requiring optimization.

Business Implication:

These insights support pricing optimization, product portfolio management, and profitability improvement initiatives.

Objective 4: Leverage Reviews, Ratings, and Customer Sentiment to Enhance Conversion and Retention

Key Insights:

- Review score analysis indicates generally positive customer feedback across the platform.
- Product categories differ considerably in average customer ratings.
- Higher customer satisfaction is associated with stronger customer retention potential.
- Categories receiving consistently positive reviews demonstrate stronger customer engagement.
- Review patterns provide valuable indicators of product and service quality.

Business Implication:

Customer feedback can be used to improve product offerings, strengthen brand perception, and increase customer loyalty.

Objective 5: Evaluate Logistics and Operational Efficiency to Improve Customer Experience

Key Insights:

- Delivery performance varies across orders and customer locations.
- Delivery delays have a noticeable impact on customer experience and satisfaction.
- On-time deliveries tend to receive better customer ratings than delayed deliveries.
- Operational analysis identified opportunities to improve fulfillment efficiency.
- Shipping performance plays a critical role in maintaining customer satisfaction levels.

Business Implication:

Improving delivery efficiency can enhance customer experience, reduce dissatisfaction, and strengthen retention.

Objective 6: Optimize Customer Retention and Lifetime Value Across Key Product Categories

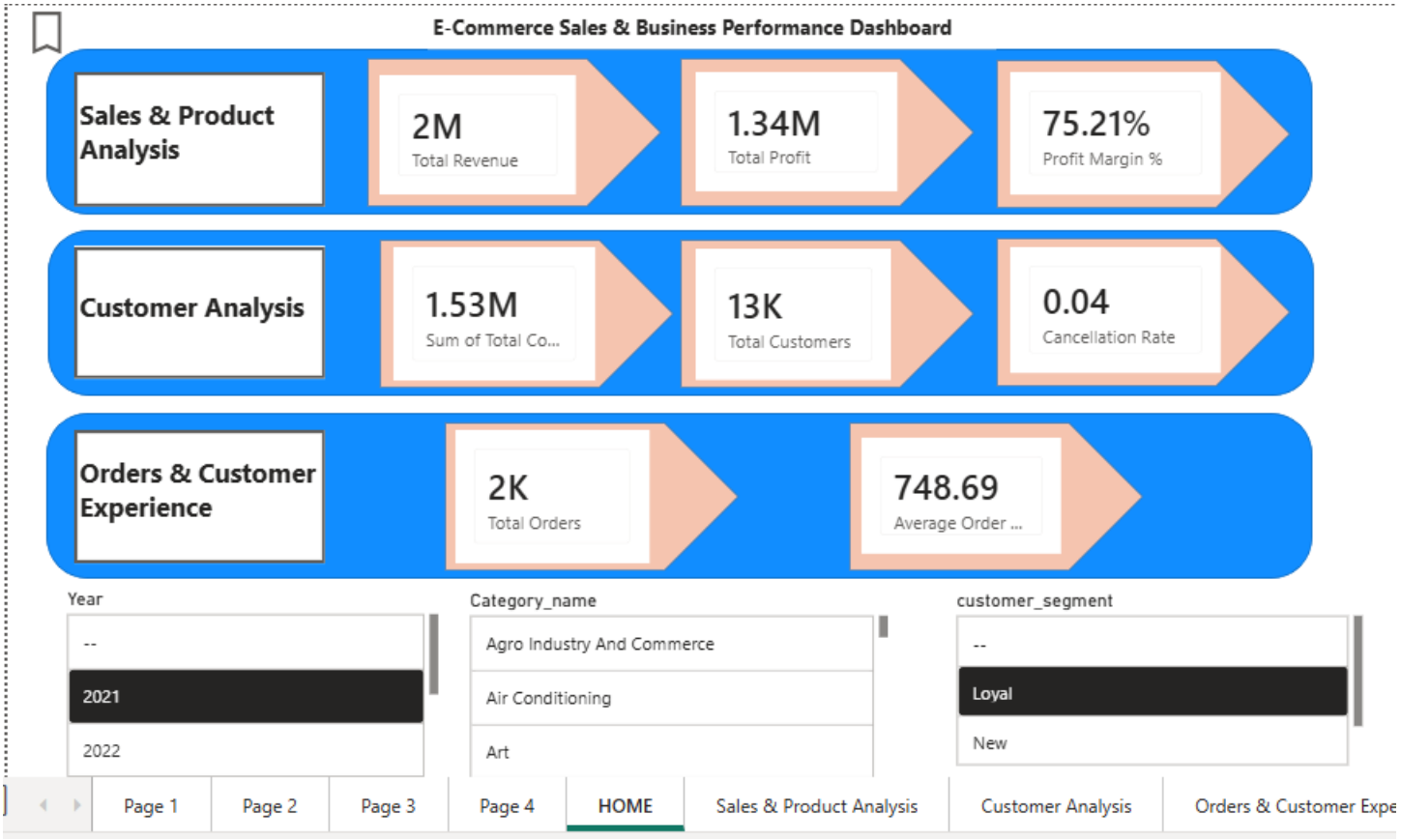
Key Insights:

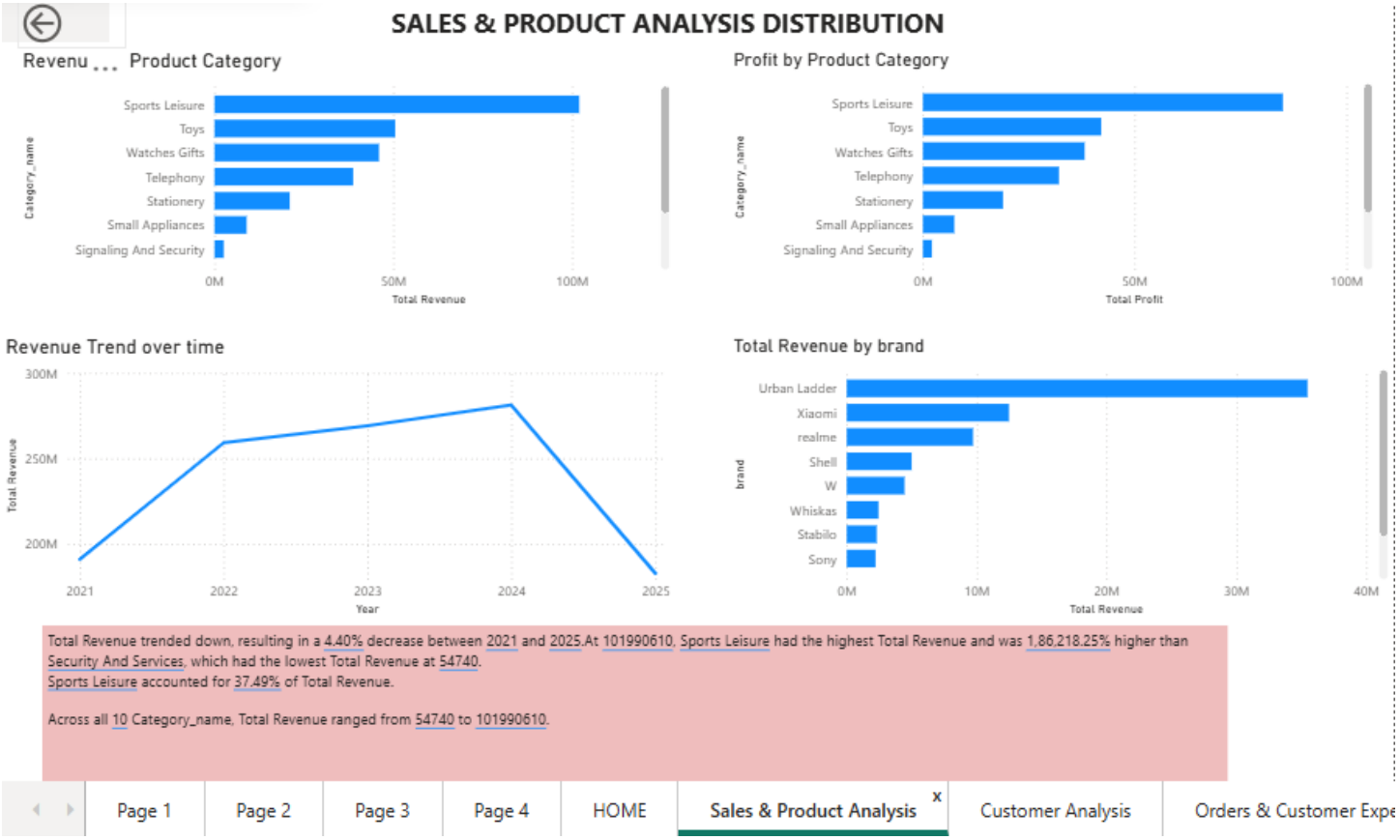
- Loyal customers generate substantially higher revenue than one-time customers.
- Customer lifetime value varies significantly across customer segments.
- Repeat purchase behavior contributes considerably to long-term business performance.
- Delivery quality, customer satisfaction, and product experience influence retention outcomes.
- Customer engagement patterns suggest opportunities for targeted retention campaigns and loyalty programs.

Business Implication:

Focusing on customer retention and lifetime value can deliver sustainable revenue growth and improve long-term profitability.

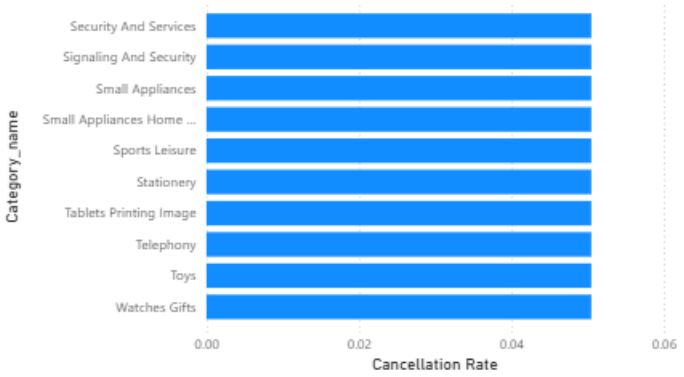
15. Power Dashboard:





CUSTOMER ANALYSIS OVERVIEW

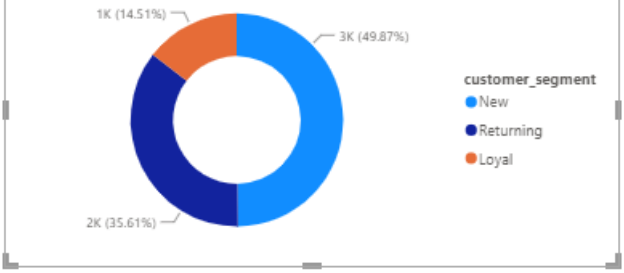
Cancellation Rate by Category_name



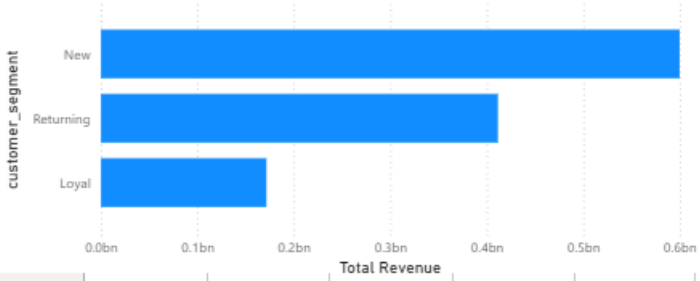
Category_name

- Agro Industry And Commerce
- Air Conditioning
- Art

Returned Orders by customer_segment



Total Revenue by customer_segment



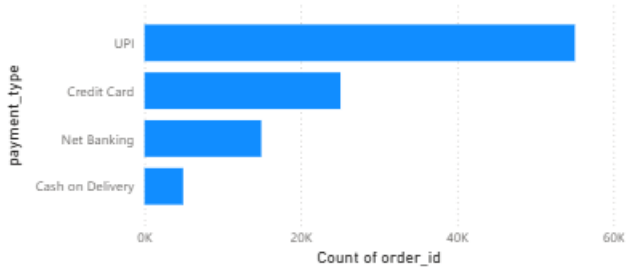
All Category_name had Cancellation Rate of 0.05.

Across all 10 Category_name, Cancellation Rate ranged from 0.05 to 0.05.

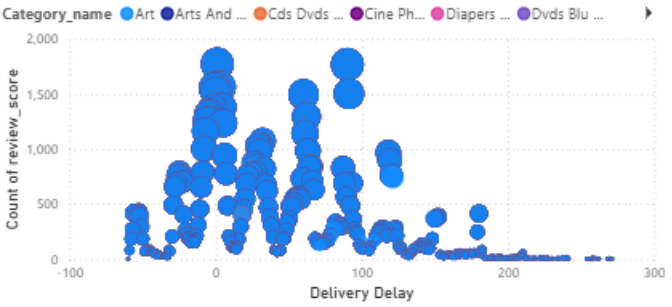
New accounted for 50.71% of Total Revenue.

ORDERS & CUSTOMER EXPERIENCE SCORECARD

Count of order_id by payment_type



Count of review_score and Total Orders by Category_name and Delivery Delay



Count of order_id by review_score and order_status



At 55,055, UPI had the highest Count of order_id and was 1,015.38% higher than Cash on Delivery, which had the lowest Count of order_id at 4,936.

UPI had the highest Count of order_id at 55,055, followed by Credit Card, Net Banking, and Cash on Delivery.

UPI accounted for 55.06% of Count of order_id.

Across all 4 payment_type, Count of order_id ranged from 4,936 to 55,055.

delivered had the highest total Count of order_id at 4,24,855, followed by returned at 35000 and cancelled at 25240.
delivered had the highest average Count of order_id at 84971, followed by returned at 7000 and cancelled at 5048.

Most Delivery Delay were below 320 in Count of review_score and below 330 in Total Orders.

16. Statistical Insights:

Hypothesis	Test	Statistic	p-value	Decision	Interpretation
H1 Delivery Delay → Cancellation	Two-Proportion Z-Test	z=-103.1500	p=1.0000	Fail to Reject H0	Implemented test does not support higher cancellation for delayed orders.
H2 Discount → Profit	Welch T-Test	t=89.9065	p≈0	Reject H0	High-discount mean profit is negative; low/moderate discount mean profit is positive.
H3 Delivery Delay → Satisfaction	Independent T-Test	t=0.9038	p=0.3661	Fail to Reject H0	No significant difference in mean review score between delayed and on-time orders.
H4 Repeat → Revenue	Descriptive comparison	—	—	Descriptive	Repeat customers have substantially higher average and total revenue.
H5 Category → Sales	Category-wise analysis	—	—	Descriptive	Sales differ across categories; furniture_decor leads.
H6 Category → Profitability	Category-wise analysis	—	—	Descriptive	Profitability differs across categories; furniture_decor leads.
H7 Higher Discount → Lower Profit	Correlation/group analysis	—	—	Supported by evidence	High discounts are associated with materially lower profit.
H8 Review → Repeat Purchase	Group/ML analysis	—	—	Weak	Simple regression coefficient -0.0039; test $R^2 \approx -0.0002$.
H9 Delivery → Review	Correlation/ML	—	—	Weak/inconclusive	Correlation≈0.0030; simple model provides no useful predictive signal.
H10 Loyalty → Revenue	Welch T-Test	t=81.3509	p≈0	Reject H0	Loyal customers have significantly higher revenue.

17.Machine Learning:

Model	X → Y	Obs.	Coefficient	Test R^2	MAE	RMSE
H2 Price → Review	unit_price → review_score	145,585	-0.0028	-0.0001	1.0458	1.3364
H3 Delay → Review	Delivery Delay → review_score	91,971	0.0061	-0.0001	1.0370	1.3297
H4 Fulfillment → Review	Fulfillment Time → review_score	24,101	0.0007	-0.0003	1.0770	1.3709
H5 Delay → Churn	Delivery Delay → churn	49,745	-0.0028	0.0001	0.4988	0.4994
H6 Review → Reorder	avg_review_score → reorder_rate	51,131	-0.0039	-0.0002	0.4998	0.4999
H7 Loyalty → LTV	loyalty_score → lifetime_revenue	41,419	14,359.0126	0.2738	12,949.90	22,246.79
H8 Discount → AOV	discount_rate → AOV	—	7,614.5117	0.1653	8,702.39	15,773.29
H9 Delay → Cancellation	Delivery Delay → cancelled	—	-0.0000	1.0000	0	0
H10 Discount → Re-engagement	avg_discount → reengagement	—	-0.0034	-0.0002	0.4836	0.4914

18.Recommendations Development:

Priority	Recommendation	KPI
1	Convert one-time customers into second-time buyers using targeted follow-ups.	2nd-purchase rate; 30/60/90-day repeat rate
2	Protect furniture_decor's sales/profit leadership and identify adjacent high-margin categories.	Revenue; profit; margin
3	Avoid blanket high discounts; test controlled discount bands.	Profit/order; AOV
4	Use separate retention journeys for New and Loyal customers.	Repeat rate; LTV
5	Investigate high-cancellation/high-return categories, considering sample size.	Cancellation; return rate
6	Monitor delivery-delay outliers and operational causes.	Average delay; on-time rate
7	Use review score as one input in customer-experience management.	Review score; repeat rate
8	Govern KPI definitions across SQL, Excel, Python and Power BI.	KPI consistency

19.Limitations & Challenges:

1. True funnel conversion is unavailable because sessions and add-to-cart events are absent.
2. Excel KPI formulas require reconciliation with Python/SQL definitions.
3. Delivery delay has high dispersion; median and distribution should accompany the mean.
4. Small category denominators can make cancellation/return rates unstable.
5. Most simple regression models have near-zero R^2.
6. Historical relationships are observational and do not prove causality.

20.Future Scope & Next Steps:

- Integrate impressions, product views, add-to-cart, checkout and purchase events.
- Develop cohort retention, churn and customer-lifetime-value models.
- Build personalized recommendations from purchase history and category affinity.
- Use A/B testing for discounts, loyalty incentives and re-engagement.
- Add review-text sentiment analysis.
- Create delivery-risk monitoring by region and category.
- Govern KPI definitions across SQL, Excel, Python and Power BI.
- Use multivariate/nonlinear models for future predictive deployment.

21.Final Management Conclusion:

Customer loyalty is the clearest high-value growth lever: repeat customers average about twice the revenue of one-time customers. furniture_decor leads both sales and profit, while high discounting is associated with materially lower profit.

The recommended strategy is to protect profitable categories, convert one-time customers into repeat customers, control discount leakage, investigate high-risk categories and delivery issues, and maintain one governed KPI layer across all analytical tools.

22.Project Deliverables Used:

Deliverable	**Role**
Project.ipynb	Python loading, cleaning, EDA, KPI calculation and business insights.
Project.sql	MySQL validation, master analytical view, business queries and dashboard result sets.
Dashboard.xlsx	Excel/pivot analyses for revenue, order volume, repeat customers, reviews and KPI questions.
Project.pbix	Interactive Power BI model/dashboard.
Project (Statistical)	H1–H10 hypothesis validation, Z/T tests and final validation.
Project(ML).ipynb	Simple linear regression and model evaluation.

23. Conclusions:

This project successfully analyzed e-commerce sales data to identify the key factors influencing customer conversion, retention, and overall business performance. Using Python, SQL, Excel, and Power BI, data was cleaned, transformed, and analyzed to uncover meaningful patterns in customer behaviour, product performance, sales trends, and operational efficiency.The analysis revealed that loyal customers contribute significantly higher revenue than one-time customers, highlighting the importance of customer retention strategies. Product category performance, seasonal sales patterns, customer reviews, and delivery experience were found to have a substantial impact on sales outcomes and customer satisfaction. Interactive dashboards and statistical analysis provided valuable insights for monitoring business performance and supporting data-driven decision-making.Overall, the project demonstrates how data analytics can transform raw e-commerce data into actionable business intelligence. The insights and recommendations generated can help improve customer conversion, increase repeat purchases, optimize operational processes, and drive sustainable business growth.